

3a) $T = \frac{2\pi}{\omega_0} = 1$ $\omega_0 = 2\pi \text{ rad/s}$

$$\frac{2\pi}{\sqrt{\omega_0^2 - \beta^2}} = 1.001 \quad \sqrt{(2\pi)^2 - \beta^2} = \frac{2\pi}{1.001}$$

$$\beta^2 = (2\pi)^2 - \left(\frac{2\pi}{1.001}\right)^2 \quad \beta = \sqrt{(2\pi)^2 - \left(\frac{2\pi}{1.001}\right)^2} = 2\pi \sqrt{1 - \frac{1}{1.001^2}} = 0.2808$$

3b) $A = A_0 e^{-\beta t}$ $\frac{A}{A_0} = e^{-\beta t}$

$$e^{-0.2808 \times 10 \times 1.001}$$

$$0.0602 \times 100$$

$$= 6.02\%$$

$$e = 0.0602$$

The amplitude is 6.02% of the original amplitude after 10 periods

Decreases by 94% from the original amplitude.

3c) The change in amplitude is more noticeable since the amplitude decreases rapidly with more oscillations.

The change in period is too small to notice (0.01s) and stays constant along the oscillations

3d) $e^{-\beta T_1} = \frac{1}{2}$ $e^{\beta T_1} = 2$ $\beta T_1 = \ln(2)$

$$\frac{2\pi\beta}{\sqrt{\omega_0^2 - \beta^2}} = \frac{2\pi\beta}{\sqrt{(2\pi)^2 - \beta^2}} = \ln(2)$$

$$(2\pi)^2 \beta^2 = \ln(2)^2 ((2\pi)^2 - \beta^2)$$

$$((2\pi)^2 + \ln(2)^2) \beta^2 = (2\pi)^2 \ln(2)^2$$

$$\beta = \frac{2\pi \ln(2)}{\sqrt{(2\pi)^2 + \ln(2)^2}} = 0.6895$$

3e) $T = \frac{2\pi}{\sqrt{(2\pi)^2 - (0.689)^2}} = 1.006s$

$$3f) \frac{A}{A_0} = e^{-0.689 \times 10 \times 1.006} = 9.762 \times 10^{-4} \quad 9.762 \times 10^{-4} \times 100 = 0.09762\%$$

The amplitude is 0.09762% of the original amplitude after 10 periods.
Decreases by 99.902% from the original amplitude

3g) The change in amplitude is more noticeable since the amplitude decreases significantly after a few oscillations (50% in one period)

The change in period is 0.006s which cannot be noticed easily when viewing the oscillations visually.